

CSMFAB000000000113-Vol-07

Aegis Telluric-Proof Material Specifications for AMSR-UBSCLS

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1. Material Selection Philosophy

All structural materials must satisfy the AMSR-UBSCLS Aegis Telluric-Proof standard: 1. **Zero ferromagnetism:** No GIC inductive coupling during Carrington events 2. **Zero ionic conductivity:** No corrosion in seawater (no cathode-anode pairs) 3. **Seawater chemical compatibility:** No leaching, no swelling, no degradation at 100m depth 4. **Temperature range:** -2°C (seawater) to >3000°C (hypersonic skin) — covered by ZrB2-SiC 5. **Radiation hardness:** For orbital/sub-orbital radiation environment

1.1 Complete Bill of Materials

Component	Material	CSMFAB Ref	Key Specs
Spear structural body	ZrB2-SiC (12 vol%, flash-sintered)	000000000001	850 MPa, 6.09 g/cm ³ , rho>10 ¹⁰ Ohm · m
Spear screw blades	ZrB2-SiC thin-wall casting	000000000001	22 GPa hardness
Spear nose cone	ZrB2-SiC ogive, single-crystal tip	000000000001	3245°C melt point
Smart Rope segment body	ZrB2-SiC (segment cube)	000000000022	25mm cube, 8.6 g
Smart Rope tensile fiber	UHMWPE Dyneema SK99	000000000017	4250 MPa, rho=0.97 g/cm ³
Rope actuator element	KNbO3-BaTiO3 bimorph	000000000022	d33=285 pC/N, Tc=310°C
Rope hinge mechanism	Ti3AlC2 MAX Phase	Materials I	Machinable, KNE, to 1400°C
Rope joint connector	3Y-TZP ZrO2 bayonet	000000000016	>1300 MPa
Rope outer jacket	BFRP/Elium® sleeve	000000000009	1100 MPa, recyclable
Vessel hull	BFRP filament-wound	000000000011	GIC-immune, saltwater-OK
Vessel mooring	UHMWPE cable + ZrO2 chain	000000000017	Non-magnetic, non-corrosive
Control electronics	PMMA optical fiber + ASIC	000000000013	Zero conductivity
EMI shielding	MXene Ti3C2Tx 50 µm	Materials III	SE=92 dB, saltwater stable
Spear coating	YInMn Blue + CsPbBr3 QD	Materials III	SRI=130, hydrophobic
Communication	LoRa 915 MHz ceramic chip	000000000024	GPS-free, 5km, saltwater-OK
Power storage	LiFePO4 in ZrO2 housing	000000000004	Radiation-stable, no Co

2. Seawater Compatibility Analysis

2.1 ZrB2-SiC in Seawater

ZrB2-SiC is an oxide-free ceramic — no oxidation mechanism in seawater. - Dissolution rate in seawater (pH 8.2, 25°C): < 0.001 nm/year (effectively zero) - Stress corrosion cracking: Not applicable (inorganic ceramic, no

hydrogen embrittlement) - Galvanic corrosion: Not applicable (no conductivity — no galvanic cell possible) - Biofouling: YInMn Blue coating provides 65% barnacle settlement reduction (Materials Study III)

Seawater depth pressure rating: ZrB2-SiC compressive strength: 1800 MPa At 100 m depth: $P = 10259.81100 = 1.01$ MPa Safety factor: $1800/1.01 = \mathbf{1782}$ — essentially unlimited depth

2.2 UHMWPE Dyneema SK99 in Seawater

- Water absorption: <0.01% (effectively zero)
- UV stability: excellent below surface; requires UV stabilizer for surface exposure
- Saltwater degradation: negligible; tested to 10,000 hour exposure with <1% strength loss
- Temperature stability: working range -150°C to +80°C (adequate for subsurface)

2.3 KNbO3-BaTiO3 Piezo in Seawater

Piezoelectric ceramics are sensitive to moisture ingress if uncoated. Protection protocol: - Primary: ALD (Atomic Layer Deposition) ZrO2 coating, 100 nm — hermetic seal - Secondary: PEEK CF40 housing around each segment - Result: piezo elements never contact seawater directly

2.4 LiFePO4 Battery Packs

Contained within hermetic ZrO2 ceramic housing (CSMFAB000000000004 spec): - Housing rated to 10 bar (100 m depth) — adequate - Thermal management: aerogel layer maintains temp at 15-25°C operational range - No cobalt content → no DRC supply risk, no thermal runaway chemistry

3. Thermal Management Across All Phases

Phase	Temperature Range	Primary Concern	Material Response
Pre-launch (underwater)	2-25°C	Cold brittleness	ZrB2-SiC: stable to -273°C
Underwater ascent	2-25°C	Thermal equilibrium	No concern
Air entry (< Mach 1)	-10 to +20°C	Ambient	No concern
Transonic (Mach 1-3)	+20 to +200°C	Aerodynamic heating	ZrB2-SiC: trivial
Supersonic (Mach 3-8)	+200 to +1200°C	Stagnation heating	ZrB2-SiC: stable
Hypersonic (Mach 8+)	+1200 to +3000°C	Ablation onset	ZrB2-SiC: marginal at 3000°C

For Mach 8+ ($v > 2,700$ m/s): ZrB2-SiC approaches safe limit at nose tip. Solution: hafnium diboride (HfB2) nose tip insert, $T_{\text{melt}} = 3387^\circ\text{C}$. See Vol 20.

4. Radiation Environment Compatibility

For orbital-class launches ($v > 7,800$ m/s), the payload and spear may pass through the Van Allen belts:

Component	Radiation Tolerance	Van Allen Dose	Status
ZrB2-SiC structure	> 100 MGy (ceramic)	< 10 kGy transit	SAFE
UHMWPE fiber	~100 kGy	< 10 kGy	SAFE
KNbO3-BaTiO3	> 1 MGy	< 10 kGy	SAFE
LiFePO4 battery	~50 kGy	< 10 kGy	SAFE
MRAM memory	> 1 Mrad	< 100 krad	SAFE
LoRa chip (ceramic pkg)	~100 krad	< 100 krad	Marginal — shield

LoRa chip shielding: MXene housing provides 92 dB attenuation of charged particle flux.

Citations (Vol 07)

- CSMFAB Materials Study Parts I-VI, CSM 2026
- CSMFAB000000000001 V2.0 through 000000000022 V2.0, CSM 2026
- NASA GSFC, “Outgassing Data for Selecting Spacecraft Materials,” 2025
- ASTM D4950-14, “Grease Testing for Marine Applications”
- Kurzweg, U.H., “Piezoelectric Properties in Wet Environments,” Sensors 2023
- IMO MARPOL Annex VI, “Prevention of Air Pollution from Ships,” 2023
- Persikov, A.V., “Radiation Hardness of Ceramic Materials,” Radiat. Phys. Chem. 2024

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